Working principle of centrifugal pump

The impeller starts rotating after the pump is filled with water and is run by a generator or a steam engine as shown in figure 4.27. When the impeller rotates, due to the centrifugal force, low pressure will be produced. Due to the low pressure, water is sucked in.

The impeller converts the mechanical power into rotating force. Due to conversion of rotating force into pressure force, the water is delivered with the help of water pump box. The amount of energy on the surface of the water is directly proportional to the velocity of the impeller.

If the speed and size of the impeller is high, the force produced in the water will be high. When the water delivers from impeller, primarily it creates friction in the pump box. Secondly, the speed of water is converted into pressure force due to the friction produced in the delivery side. Hence, the pressure of the water is equivalent to the speed of rotation by the impeller.

a. Friction power

When the water passes through the pipe, it creates friction inside the pipe. The friction produced inside the pipe will be according to the speed of the water. Due to this, more power is required to pump the water up. Because of friction produced inside the water pipe, power loss is occurred. The power used to compensate this is called as friction force. It will be good, if the length of the suction pipe is less. The flow of water in higher diameter produces less friction.

b. Suction power

Suction conditions are some of the factors which affect the centrifugal pump operation. A pump cannot pull or "suck" water up into suction pipe, because water does not have tensile strength. When a pump creates suction, it is simply reducing local pressure by creating a partial vacuum (Sucks out the air above the water). External pressure acting on the surface of the liquid pushes the liquid up the suction pipe into the pump.

i. Static suction head

The static suction head refers to the vertical height of the water absorbed in well or the horizontal center at the water pump from the water level of the underground tank. Suction head does not depend upon the length of the pipe. It is from the water level to the pump centre and not from the foot valve or the bottom of the well.

ii. Static delivery head

The static delivery head indicates the vertical height from the horizontal line of the water to the water delivered to the water tank. This does not indicate the length of the delivery pipe.

iii. Priming

When the suction pipe and pump is filled with water, the air inside the pipe should be removed. This method is said

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